

Eighty760 App Summary

What It Is

Eighty760 is a web-based simulation framework for modeling 24/7 carbon-free energy portfolios and hourly matching across a full year (8,760 hours). It acts as a digital twin for renewable portfolios to compare generation against real-world load profiles.

Who It's For

Primary users are energy portfolio planners, sustainability teams, and analysts evaluating 24/7 CFE performance.

What It Does

- Interactive portfolio simulator to adjust solar, wind, and battery capacity in real time.
- Aggregation tool with live market data, detailed analysis, financials, and scenario comparison.
- Scenario comparison to save and evaluate multiple portfolio configurations side by side.
- Financial modeling for PPA settlement costs, basis risk, battery arbitrage revenue, and REC costs.
- Weather performance backcasting using historical data (2020-2026).
- Global real-time tickers and hourly generation profiles (duck curve, net load).

How It Works (Repo Evidence)

- Next.js 14 App Router UI in `app/` with page routes for home, aggregation, weather performance, and whitepaper.
- React components in `components/` and core simulation UI in `components/Simulator.tsx` with supporting logic in `lib/aggregation/`.
- API routes under `app/api/` for market data, news, prices, and tickers; external API clients in `lib/external/`.
- Python utilities in `scripts/` for historical price and weather data processing.
- Persistent database or backend service beyond Next.js API routes: Not found in repo.

How To Run (Minimal)

- Prerequisite: Node.js v18 or higher.
- Install dependencies: `npm install`
- Start dev server: `npm run dev`
- Open: `http://localhost:3000`